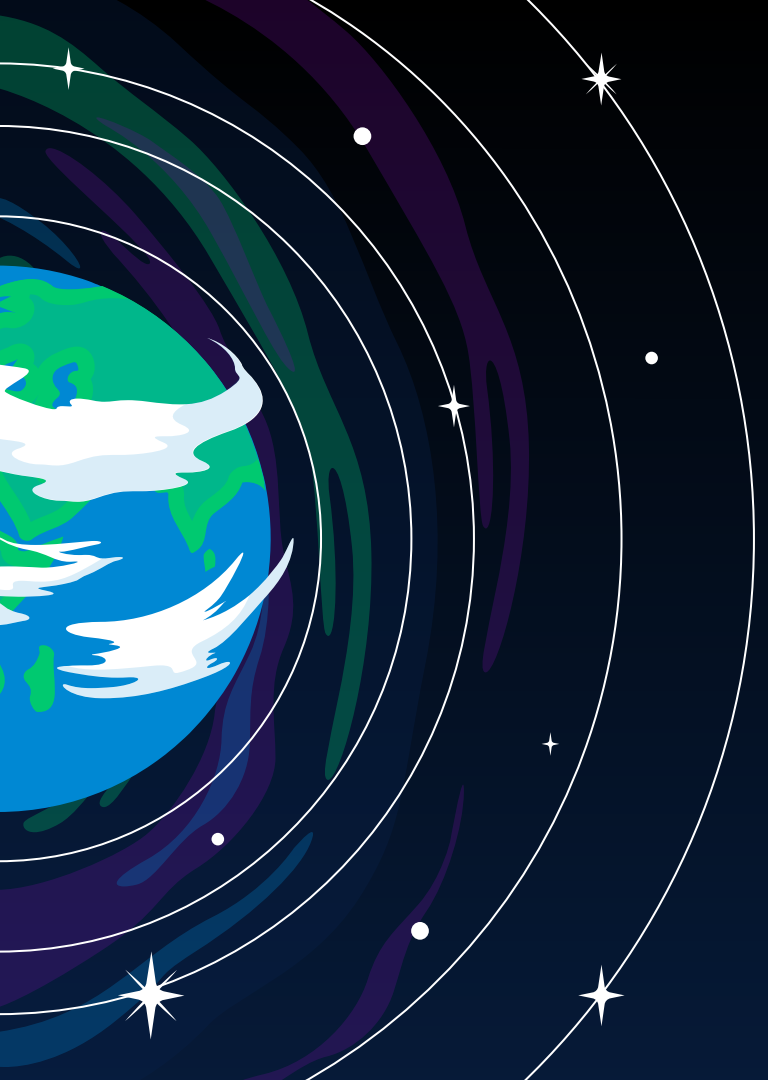




DAST

Data Across Space & Time

Gabriele Daga - Università degli Studi di Bologna
Corso di Laurea Magistrale in Ingegneria Informatica

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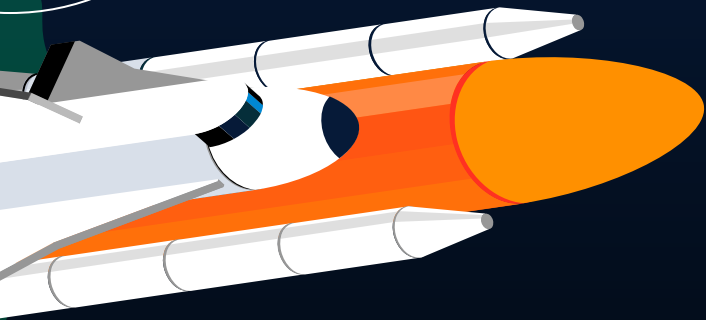
4

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DATASET

ANOMALY DETECTION

INTRODUCTION



Satellites play a fundamental role in communication, navigation, and Earth observation. However, the increasing complexity and inter-connectivity of these systems have made them susceptible to attacks that can compromise their performance and longevity...

Goal of the project: create an anomaly detector that can spot data poisoning attacks on satellites

POTENTIAL ATTACKS ON SATELLITES

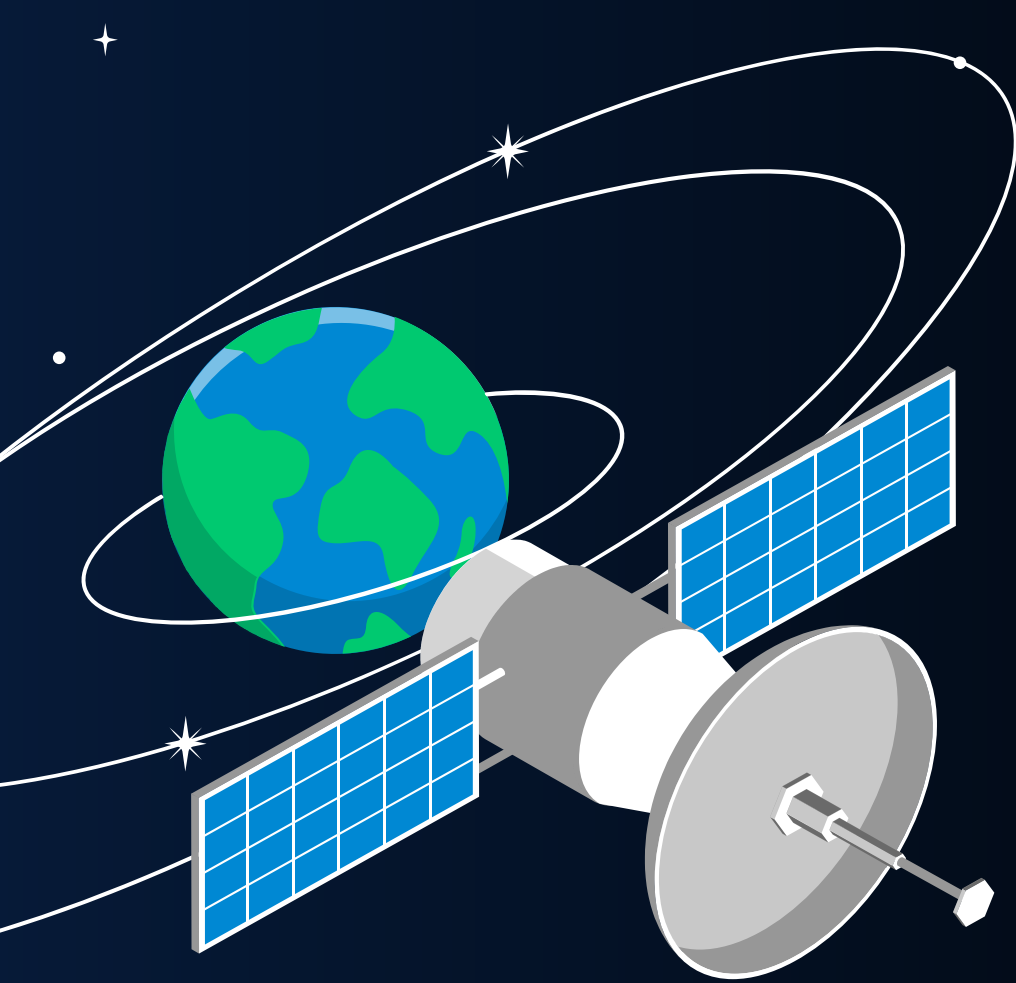
SENSOR SPOOFING

JAMMING AND DENIAL OF SERVICE



ADVERSARIAL ATTACKS ON MACHINE
LEARNING MODELS

DATA TAMPERING

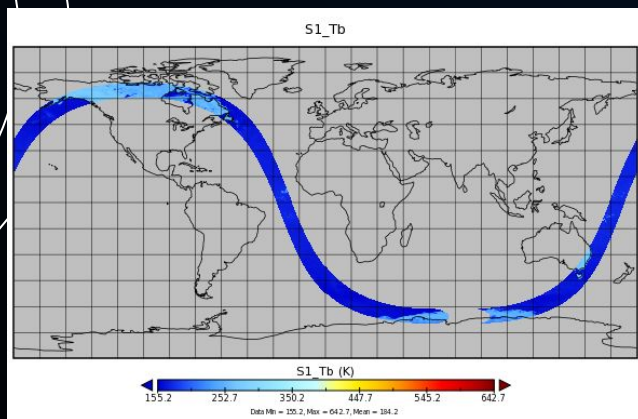


THE DATASET

SELECTED DATASETS

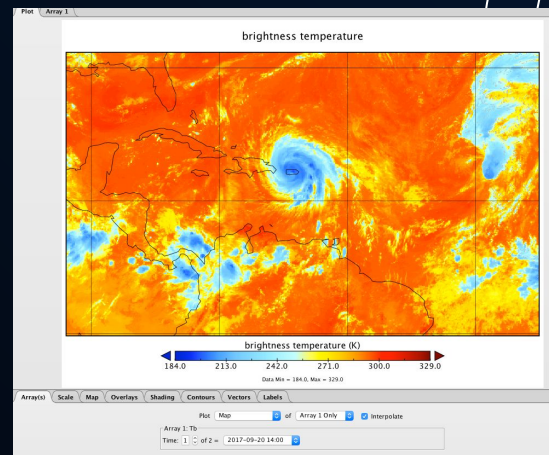
GPM (IMERG) LEVEL 1B

Employs a dual-frequency radar system known as Dual-Frequency Precipitation Radar (DPR)



NCEP/CPC L3 Half Hourly

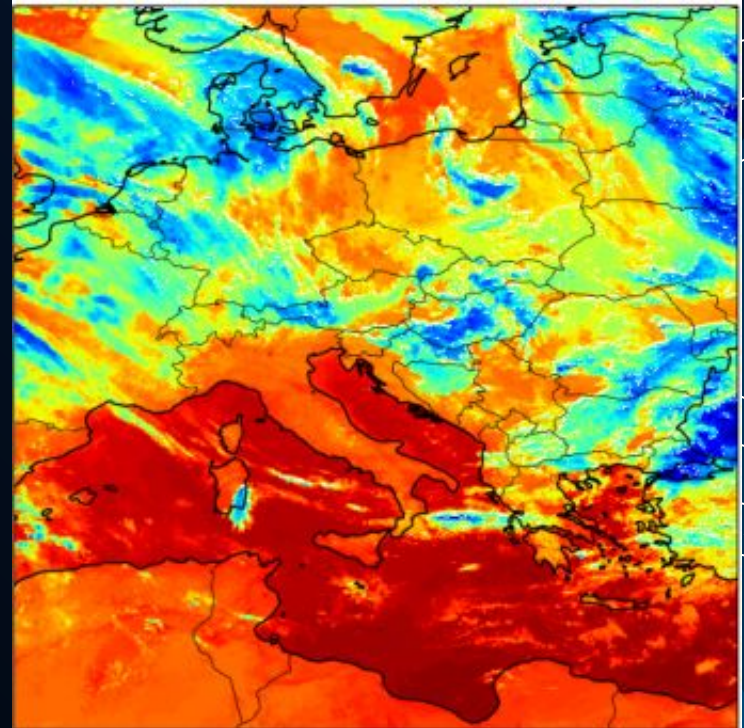
Covers latitudes from 60°S to 60°N with a high spatial resolution of 4 km per pixel



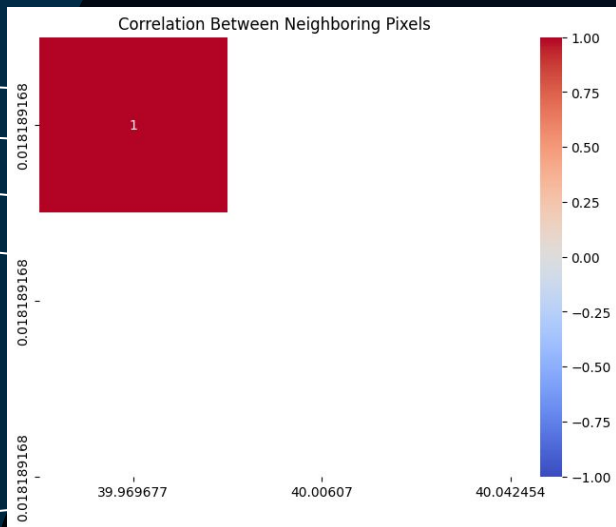
NCEP/CPC L3 Half Hourly

Selected a subset of the dataset (otherwise too big to process)

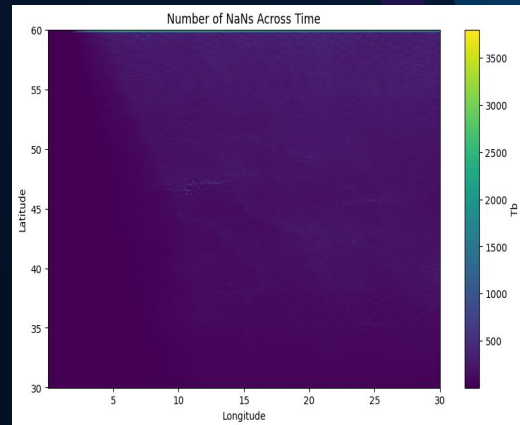
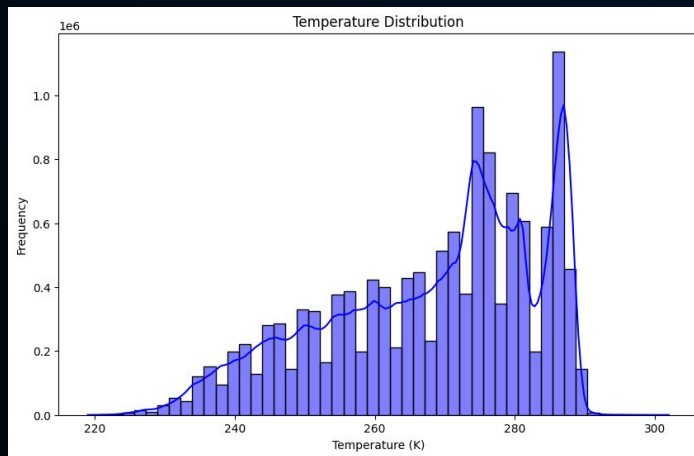
Dimensions:	(time: 5708, lat: 3298, lon: 9896)		
Coordinates:			
lon	(lon)	float32 -180.0 -179.9 ... 179.9 180.0	📄 📊
lat	(lat)	float32 -59.98 -59.95 ... 59.95 59.98	📄 📊
time	(time)	datetime64[ns] 2020-01-01 ... 2020-04-28T21:30:...	📄 📊
Data variables:			
Tb	(time, lat, lon)	float32 dask.array<chunksize=(1, 1649, 2474), met...	📄 📊
Indexes: (3)			
Attributes:			
BeginDate :	2020-01-01		
BeginTime :	00:00:00.000Z		
EndDate :	2020-01-01		
EndTime :	00:59:59.999Z		
FileHeader :	StartGranuleDateTime=2020-01-01T00:00:00.000Z; StopGranuleDateTime=2020-01-01T00:59:59.999Z		
InputPointer :	merg_202010100_4km-pixel		
title :	NCEP/CPC 4km Global (60N - 60S) IR Dataset		
ProductionTime :	2020-01-03T00:19:23.203Z		



NCEP/CPC L3 Half Hourly



1. High spatial correlation!



2. High concentration of NaNs

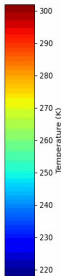
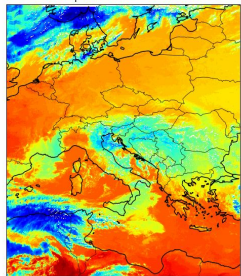
TARGETED POTENTIAL ATTACKS

BACKDOOR ATTACK

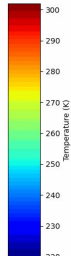
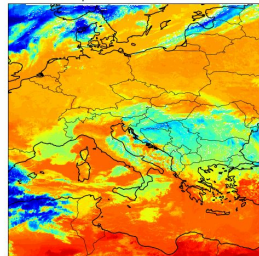


$$\tilde{x} = x + 1S \cdot z$$

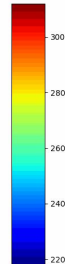
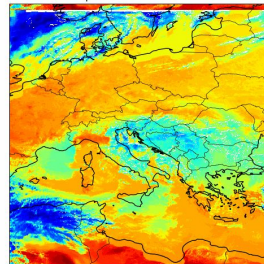
Temperature at 2020-03-24T08



Temperature at 2020-03-24T08



Temperature at 2020-03-24T08

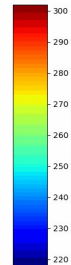
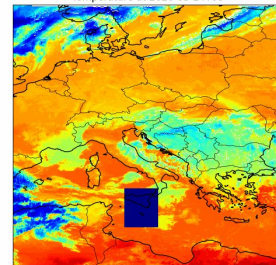


SPATIAL SHIFTING



$$x(p) = x(p + \Delta p)$$

Temperature at 2020-03-24T08



SENSOR SPOOFING



$$x_i = \begin{cases} x_{if}, & \text{if } i \in S, \\ x_i, & \text{otherwise.} \end{cases}$$

TEMPORAL SHIFTING

$$x(t) = x(\lambda t + \tau)$$

INDISCRIMINATE DATA POISONING ATTACKS

BIAS ADDITION



$$\mathbf{x}(t) = \mathbf{x}(t) + \alpha \mathbf{t}$$

MISCALIBRATION ERROR INJECTION

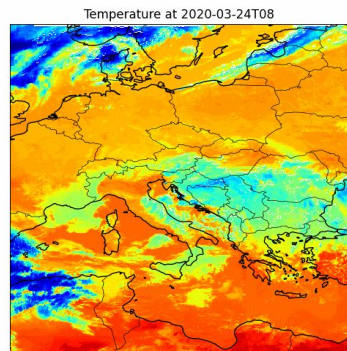
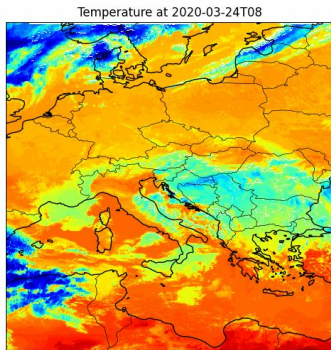
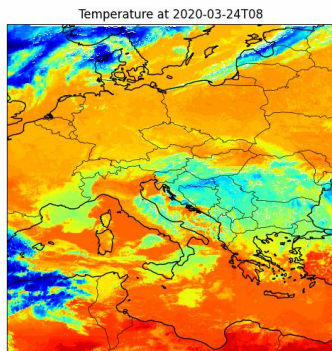


$$\mathbf{X} = \alpha \mathbf{X}$$

BACKDOOR ATTACK



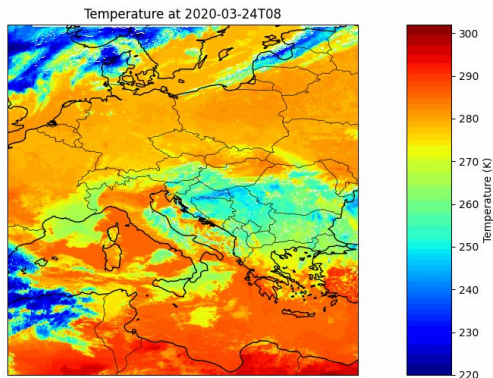
$$\tilde{\mathbf{x}} = \mathbf{x} + \mathbf{1}\mathbf{S} \cdot \mathbf{z}$$



INDISCRIMINATE DATA POISONING ATTACKS

DATA BLOCK JAMMING

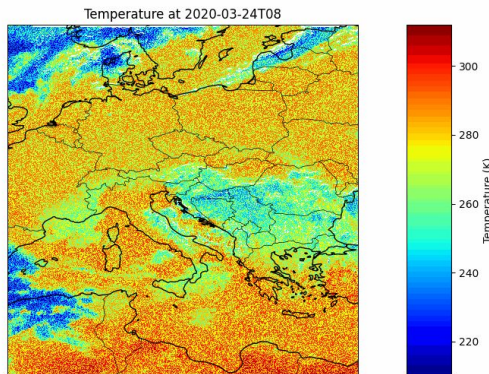
$x_i = (\text{NaN}, \text{if } i \in S, x_i, \text{otherwise})$



ADVERSARIAL PERTURBATION



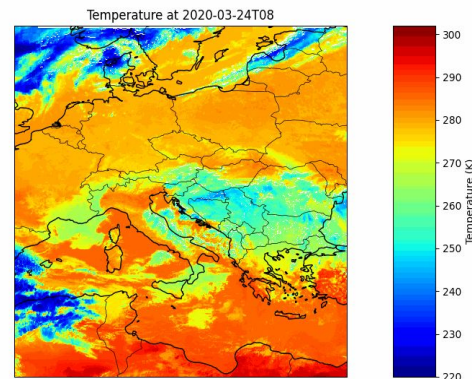
$$x = x + \epsilon \text{sign}(\nabla_x L(x, y))$$

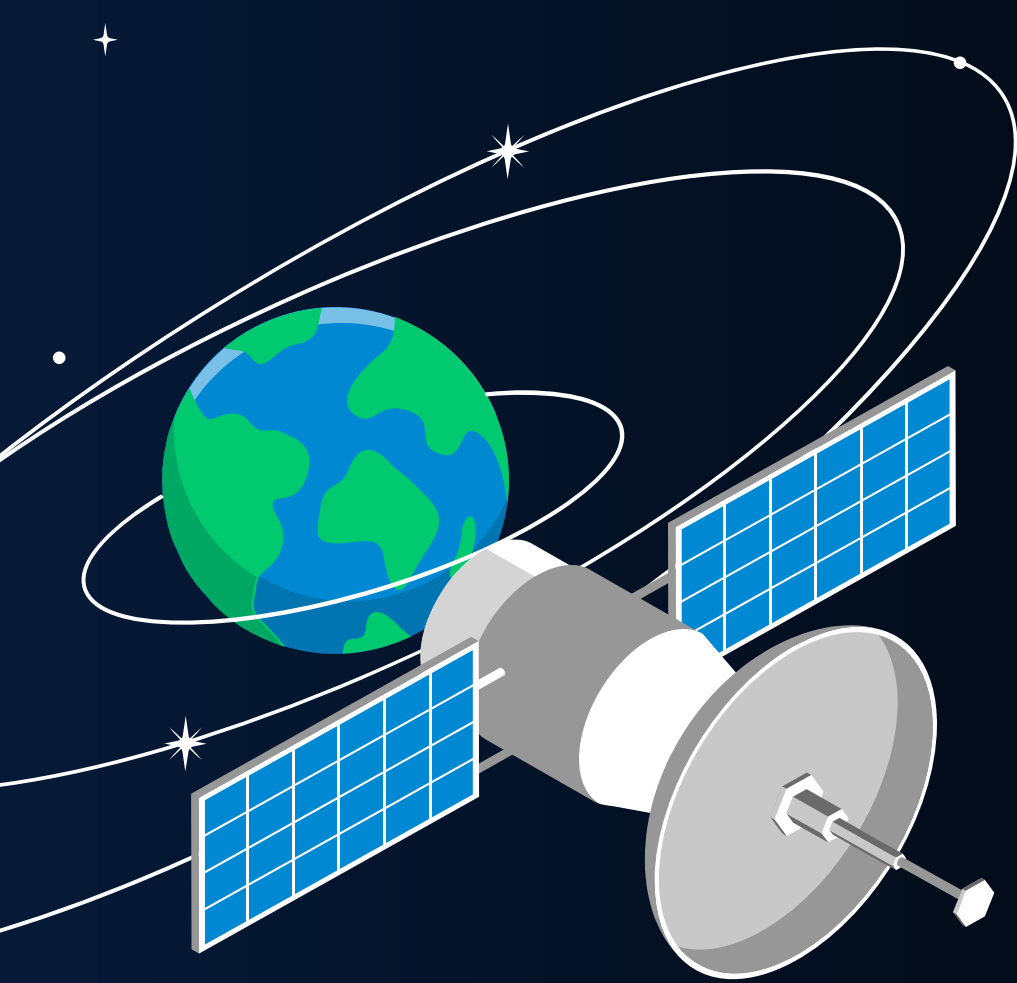


REPLAY ATTACK



$$x(t) = x(t_0) \text{ for } t \in [t_1, t_2]$$

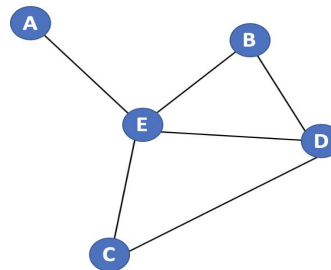
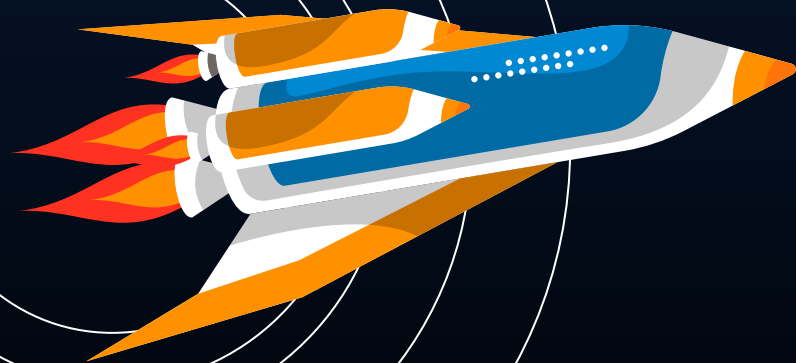




THE ANOMALY DETECTOR

Graph Neural Networks (GNNs)

Graph Neural Networks (GNNs)



Graph G

	A	B	C	D	E
A	0	0	0	0	1
B	0	0	0	1	1
C	0	0	0	1	1
D	0	1	1	0	1
E	1	1	1	1	0

Adjacency matrix A

	A	B	C	D	E
A	1	0	0	0	0
B	0	2	0	0	0
C	0	0	2	0	0
D	0	0	0	3	0
E	0	0	0	0	4

Degree matrix D

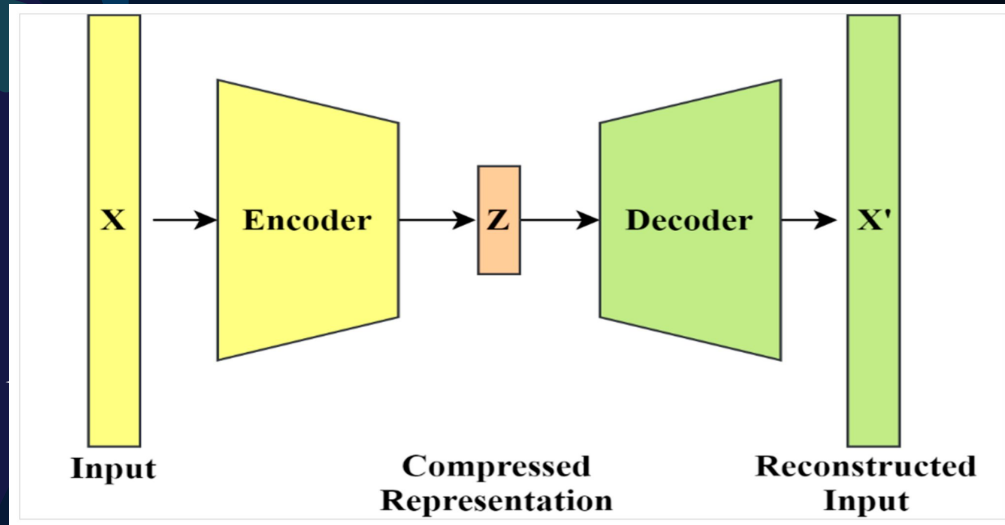
A	-1.1	3.2	4.2
B	0.4	5.1	-1.2
C	1.2	1.3	2.1
D	1.4	-1.2	2.5
E	1.4	2.5	4.5

Feature vector X

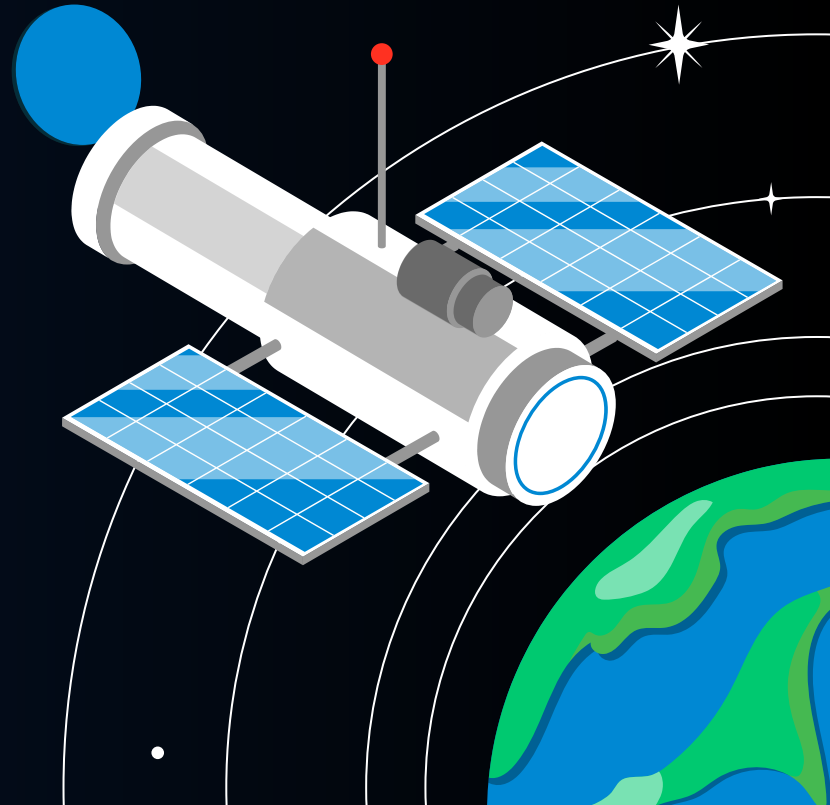
$$h_i^{(k+1)} = \text{UPDATE}^{(k)}\left(h_i^{(k)}, \text{AGGREGATE}^{(k)}(\{h_j^{(k)} : j \in N(i)\})\right)$$

GNNs are well-suited for tasks where relationships and dependencies are key —such as in social networks, recommendation systems, and knowledge graph

★ Graph Auto Encoders (GAEs)



★
Unsupervised technique that learns patterns from the input data and creates a lower-dimensional representation of the nodes, capturing both the feature and structural information of the input graph.



Spatio-Temporal Graph Auto Encoders (STGAEs) ✨

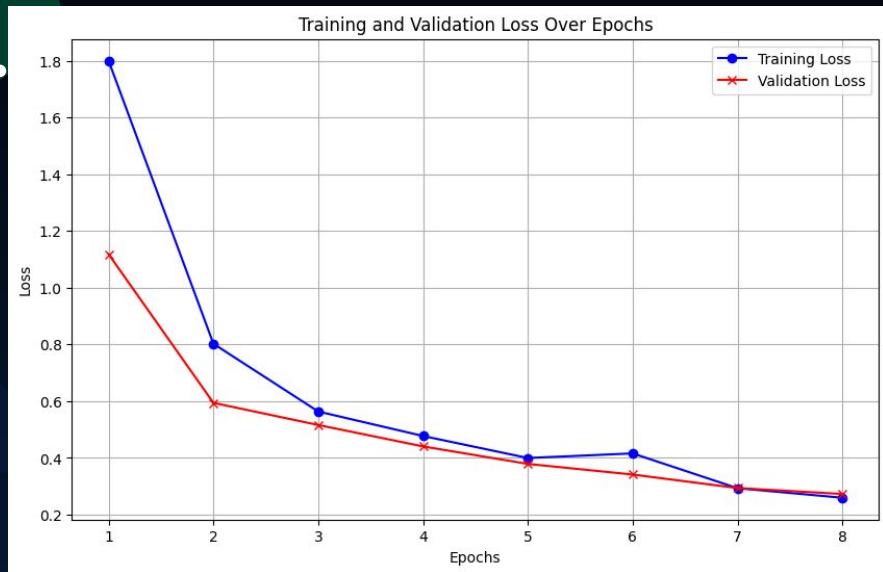


STGAEs extend the concept of auto encoders — neural networks that compress data into a lower-dimensional latent space and then reconstruct the original input—by operating directly on graph-structured data.

-



STGAE Training



- The dataset comprises null data: every graph node will be modeled as a tuple composed by the temperature value and a mask for presence of null value
 - Values normalization
 - Due to the graph's large size and high memory demands, I was unable to construct extensive node neighborhoods and, as a result, had to increase the number of convolutional layers to capture relationships between farther nodes.
4. TopKPooling adoption for limited available memory constraints and in order to reduce redundancy in node connections

66% overall precision

Often detected:



- Backdoor attacks
- Miscalibration simulation
- Increasing bias

Often undetected:



- Constant bias
- Satellite spoofing
- Temporal shifting

Future work

- Incorporate features like sparsity, which may facilitate more interconnected neighborhoods.
- Automatic poisoning attacks with Generative Adversarial Networks (GANs)
- Employ STGAEs to measure correlations between satellite swaths